Game Rules

Games without Chance: Combinatorial Game Theory: Game Rules

Nim

Nim is a game where players take turns removing objects from different heaps (think stacks of coins). On a players turn, the player must remove at least 1 object from one of the heaps. They can remove as many objects as they want as long as they are in the same heap. The first player to be unable to move loses.

Toads and Frogs

Toads and Frogs is a game played on a 1×n strip of squares. Every square is either empty or has a toad or a frog. Players take turns, Left moving toads to the right if the next square is an empty square and Right moving frogs to the left if the next square is empty. Also, if there is a frog in the square that a toad wants to move into, it can hop over the frog into the next square (2 squares ahead of where it started) if that square is empty. Similarly, a frog can jump over a toad to the next square if that square is empty. The first player that does not have a move loses.

Cut Cake

Cut Cake is a game played on a rectangular grid. During each players turn, they cut the rectangle into two smaller rectangles. The Left player can cut along the vertical (North to South) lines while Right can cut along the horizontal (East to West) lines. The first player without a move loses.

Hackenbush

Hackenbush is a game played on a configuration of colored lines (usually red, blue, and green). These lines are connected to the ground, either directly by touching the ground or indirectly by being connected to another line that is connected to the ground. Players take turns cutting the line segments (ie. erasing). The player Right can cut Red lines, the player Left can cut bLue lines, and both can cut green lines. When a line is cut, any remaining pieces that are no longer connected to the ground are also removed. The first player to be unable to move loses.

Run Over

Run over is a game played on a finite strip of squares, where each square is empty or occupied by a Left or Right piece. Each player can move their piece one square to the left and replace any piece that is on that square. Once pieces reach the last square, they can move off of the board.

Various Game Values

- $G = \{G^L|G^R\}$
- $-G = \{-G^R| - G^L\}$
- $G = H$ if $G - H = 0$

Equivalently, $G = H$ if $G + K$ has the same outcome in best play as $H + K$ for all games K

- $0 = \{|\}$

If $G = 0$, then the first player to move loses

- $1 = \{0|\}$

A positive game value is a Left win

- $-1 = \{|0\}$

A negative game value is a Right win

- $n = \{n - 1|\}$
- $\frac{1}{2} = \{0|1\}$
- $\star = \{0|0\}$
- $-\star = \star$
- $\star n = \{\star 0, \star 1, ..., \star n - 1| \star 0, \star 1, ..., \star n - 1\}$
- $\uparrow = \{0|\star\}$
- $\downarrow = -\uparrow$
- $G > 0$ means Left wins
- $G < 0$ means Right wins
- $G = 0$ means first player loses
- $G||0$ means second player loses

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